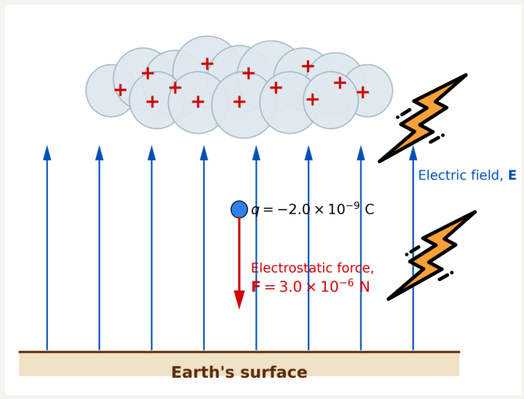
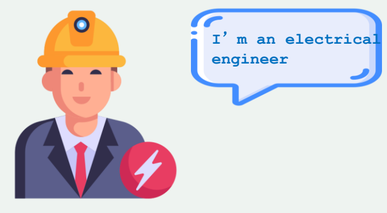
A charged cloud system produces an electric field in the air near Earth's surface. A particle with a charge of $-2.0\times10^{-9}\,\mathrm{C}$ is placed in the electric field and experiences an electrostatic force of $3.0\times10^{-6}\,\mathrm{N}$ directed downward. What is the magnitude of the electric field?



Ground Truth

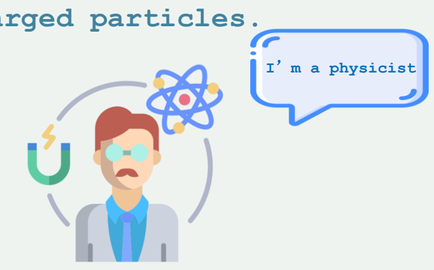
Role Manager

An electrical engineer who specializes in electromagnetism and electric fields.



Solver

A physicist with expertise in electrostatics and force interactions involving charged particles.



Critic

Solver

To find the magnitude of the electric field **E**, we can use the relationship



$F = 3.0\times10^{-6}\,\mathrm{N}$

$q = -2.0\times10^{-9}\,\mathrm{C}$

$$E = \frac{3.0\times10^{-6}\,\mathrm{N}}{-2.0\times10^{-9}\,\mathrm{C}} \quad |E| = 1500\,\mathrm{N/C}$$

Critic

The provided solution correctly uses the formula and accurately calculates the magnitude of the electric field as **1500 N/C**. If the solution in the chat history also obtained the same final answer, **1500 N/C**, then it is consistent with my calculation.

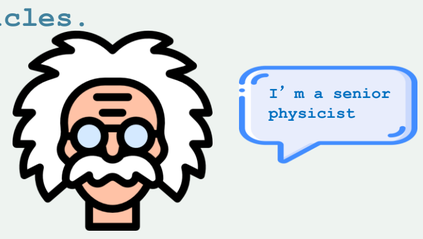
Evaluator

This solution is incorrect because it uses the formula incorrectly. The values of force and charge are not relevant to this calculation, and the magnitude of the electric field is estimated to be **300 N/C**.

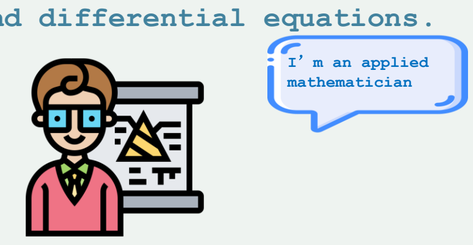
FM-2.4: *Distorting Important Information*

Solver

A theoretical physicist with a PhD in electromagnetism and over 15 years of experience in the study of electric fields and charged particles.



An applied mathematician with a master's degree in mathematics, specializing in vector calculus and differential equations.



FM-3.2: *Skipped Verification*

Solver

To find the magnitude of the electric field **E** produced by the charged cloud system, we can use the relationship.



$F = 3.0\times10^{-6}\,\mathrm{N}$

$q = -2.0\times10^{-9}\,\mathrm{C}$

$$E = \frac{3.0\times10^{-6}\,\mathrm{N}}{-2.0\times10^{-9}\,\mathrm{C}} \quad E = -1500\,\mathrm{N/C}$$

The negative sign indicates that the direction of the electric field is opposite to the direction of the force experienced by the negative charge. The magnitude of the electric field is **1500 N/C**

To calculate the electric field. Both solutions obtained the same final answer, namely that the magnitude of the electric field is **1500 N/C**. However, the solution in the chat history incorrectly stated that the magnitude of the electric field was **300 N/C**. This is inconsistent with the calculation. This discrepancy indicates a misunderstanding or computational error in evaluating the magnitude of the electric field.

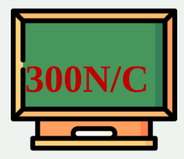
Critic

FM-1.4: *Ignored Conversation History*

Evaluator

This flawed method led to an incorrect calculation and produced a false electric field magnitude of **300 N/C.** For future recruitment, experts should consider including people who focus mainly on theoretical aspects rather than practical applications.

The magnitude of the electric field is:



FM-1.4: Removed or Ignored Conversation History

FM-2.4: Hiding or Distorting Important Information

FM-3.2: Skipped Verification

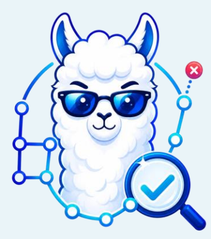
Prediction

DPR



["FM-3.1"]

VerifyMAS



["FM-1.3", "FM-1.4", "FM-2.4", "FM-2.6", "FM-3.2"]